CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

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Spring 2026
Lecture 2

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Normal

Def.
$X \sim \text{Normal}(\mu, \sigma^2)$, where $\mu \in \mathbb{R}$ and $\sigma > 0$, if X is a continuous random variable with the following probability density function (pdf):

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad \text{for all } x \in \mathbb{R}.$$

Claim:
If $X \sim \text{Normal}(\mu, \sigma^2)$ with some $\mu \in \mathbb{R}$ and $\sigma > 0$ then

1. the cumulative distribution function (cdf) is

$$F_X(x) = \Phi\left(\frac{x-\mu}{\sigma}\right), \quad \text{where } \Phi(z) \doteq \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt;$$

2. the expectation is

$$E[X] = \mu.$$

Chi-Square Distribution

Def.

Let $Z \sim N(0,1)$. Then the distribution of $U = Z^2$ is called the *chi-square* distribution with 1 *degree of freedom*:

$$U \sim \chi_1^2.$$

Chi-Square Distribution

Def.

Let $Z \sim N(0,1)$. Then the distribution of $U = Z^2$ is called the *chi-square* distribution with 1 *degree of freedom*:

$$U \sim \chi^2_1.$$

Def.

Let $U_1, U_2, \ldots, U_n \overset{\text{iid}}{\sim} \chi^2_1$. Then the distribution of $V = U_1 + U_2 + \ldots + U_n$ is called the *chi-square* distribution with n *degrees of freedom*:

$$V \sim \chi^2_n.$$

Chi-Square Distribution

Claim

If $V \sim \chi^2_n$ then $V \sim \text{Gamma}\Big(\underbrace{\frac{n}{2}}_{\alpha}, \underbrace{\frac{1}{2}}_{\lambda}\Big)$, i.e. the pdf is

$$f_V(v) = \begin{cases} \frac{1}{2^{n/2}\Gamma(n/2)} v^{(n/2)-1} e^{-v/2}, & \text{for all } v \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

t Distribution

<u>Def.</u>
Let $Z \sim N(0,1)$ and $U \sim \chi^2_n$ be independent. Then the distribution of $T = \frac{Z}{\sqrt{U/n}}$ is called the t distribution with n *degrees of freedom*:

$$T \sim t_n.$$

t Distribution

Def.
Let $Z \sim N(0,1)$ and $U \sim \chi_n^2$ be independent. Then the distribution of $T = \frac{Z}{\sqrt{U/n}}$ is called the t distribution with n *degrees of freedom*:

$$T \sim t_n.$$

Claim
If $T \sim t_n$ then the pdf of T is

$$f_T(t) = \frac{\Gamma((n+1)/2)}{\sqrt{n\pi}\,\Gamma(n/2)} \left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}} \quad \text{for all } t \in \mathbb{R}.$$

F Distribution

<u>Def.</u>
Let $U \sim \chi^2_m$ and $V \sim \chi^2_n$ be independent. Then the distribution of $W = \frac{U/m}{V/n}$ is called the F distribution with m and n *degrees of freedom*:

$$W \sim F_{m,n}.$$

F Distribution

Def.
Let $U \sim \chi^2_m$ and $V \sim \chi^2_n$ be independent. Then the distribution of $W = \frac{U/m}{V/n}$ is called the F distribution with m and n *degrees of freedom*:

$$W \sim F_{m,n}.$$

Claim
If $W \sim F_{m,n}$ then the pdf of W is

$$f_W(w) = \begin{cases} \frac{\Gamma((m+n)/2)}{\Gamma(m/2)\Gamma(n/2)} \left(\frac{m}{n}\right)^{\frac{m}{2}} w^{\frac{m}{2}-1} \left(1 + \frac{m}{n} w\right)^{-\frac{m+n}{2}}, & \text{for all } w \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

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Convergence in Probability

Def.

A sequence of random variables $X_1, X_2, X_3, \ldots$ *converges in probability* to $\alpha \in \mathbb{R}$,

$$X_n \xrightarrow{P} \alpha,$$

if for every $\varepsilon > 0$ we have

$$\lim_{n\to\infty} P(|X_n - \alpha| > \varepsilon) = 0.$$

Convergence in Distribution

<u>Def.</u>
A sequence of random variables $X_1, X_2, X_3, \ldots$ *converges in distribution* to a random variable X,

$$X_n \xrightarrow{D} X,$$

if

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

at every point at which $F_X(x)$ is continuous.
Here, $F_X(x)$ is the cdf of X and F_{X_n} is the cdf of X_n, $n = 1, 2, 3, \ldots$

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Law of Large Numbers (LLN)

Thm. (LLN)
Let $X_1, X_2, X_3, \ldots$ be a sequence of independent random variables with

$$E(X_k) = \mu \text{ and } \mathrm{Var}(X_k) = \sigma^2 \text{ for } k = 1, 2, 3, \ldots$$

Let

$$\bar{X}_n \doteq \frac{1}{n}\sum_{k=1}^{n} X_k.$$

Then, for any $\varepsilon > 0$,

$$P(|\bar{X}_n - \mu| > \varepsilon) \to 0 \text{ as } n \to \infty.$$

Central Limit Theorem (CLT)

Thm. (CLT)
Let $X_1, X_2, X_3, \ldots$ be a sequence of independent identically distributed random variables with

$$E(X_k) = \mu \text{ and } \operatorname{Var}(X_k) = \sigma^2 > 0 \text{ for } k = 1, 2, 3, \ldots$$

Let

$$\bar{X}_n \doteq \frac{1}{n} \sum_{k=1}^{n} X_k.$$

Then, assuming mgf's of X_k exist in a neighborhood of 0,

$$\lim_{n \to \infty} P\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq x \right) = \Phi(x) \text{ for all } x \in \mathbb{R},$$

where

$$\Phi(z) \doteq \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

is the cdf of a standard normal random variable.

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Definition of Sample Mean and Sample Variance

Def.

Let $X_1, X_2, \ldots, X_n$ be independent identically distributed (iid) random variables. Then we define:

- *sample mean* as

$$\bar{X} = \frac{1}{n}\sum_{k=1}^{n} X_k$$

- *sample variance* as

$$S^2 = \frac{1}{n-1}\sum_{k=1}^{n}\left(X_k - \bar{X}\right)^2$$

Distribution of Sample Mean

Claim

If $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$ then

$$\bar{X} \sim \text{Normal}\left(\mu, \frac{\sigma^2}{n}\right).$$

Independence of Sample Mean and Sample Variance

<u>Thm.</u>

If $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$ then

$$\bar{X} \;\text{ and }\; (X_1 - \bar{X},\, X_2 - \bar{X}, \ldots,\, X_n - \bar{X}) \;\text{ are independent.}$$

Independence of Sample Mean and Sample Variance

<u>Thm.</u>

If $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$ then

$$\bar{X} \text{ and } (X_1 - \bar{X},\, X_2 - \bar{X}, \ldots,\, X_n - \bar{X}) \text{ are independent.}$$

<u>Corollary</u>

If $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$ then

$$\bar{X} \text{ and } S^2 \text{ are independent.}$$

Distribution of Sample Variance

Thm.

If $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim}$ Normal(μ, σ^2) then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Distribution of Sample Variance

Thm.

If $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$ then

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Corollary

If $X_1, X_2, \ldots, X_n \overset{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$ then

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

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Hypotheses Testing: Example

Example:

Consider the following random sample from a normal distribution $N(\mu, \sigma^2)$:

$$14.2, 12.1, 15.9, 17.6, 17.4, 18.2 \quad \text{i.e.} \quad \bar{X} = 15.9 \text{ and } S = 2.3563$$

Suppose we want to test $H_0 : \mu = 12$ vs. $H_1 : \mu \neq 12$.

Hypotheses Testing: Example

Example:

Consider the following random sample from a normal distribution $N(\mu, \sigma^2)$:

$$14.2, 12.1, 15.9, 17.6, 17.4, 18.2 \quad \text{i.e.} \quad \bar{X} = 15.9 \text{ and } S = 2.3563$$

Suppose we want to test $H_0 : \mu = 12$ vs. $H_1 : \mu \neq 12$.

Then, under the H_0 hypothesis, we must have

$$\frac{\bar{X} - 12}{S/\sqrt{n}} \sim t_{n-1} \quad \text{(t-test, d.f.}=n-1)$$

In this case, we observe

1. *test statistic* is TS $\doteq \frac{\bar{X}-12}{S/\sqrt{n}} = \frac{15.9-12}{2.3563/\sqrt{6}} = 4.0542$
2. p-value $\doteq P(|T| > |\text{TS}|) = 0.0097$, where $T \sim t_{n-1}$
3. given level $\alpha = 0.05$, we observe p-value $< \alpha$, i.e. p-value is too small
   $\Rightarrow$
   reject H_0, i.e. accept H_1
   (data is statistically significant at level $\alpha = 0.05$, evidence in favor of H_1).

Testing Hypotheses about Mean

Suppose we test

$$H_0 : \mu = \mu_0 \text{ vs. } H_1 : \mu \neq \mu_0.$$

Then one can use the following *test statistics* (TS):

- TS $= \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$, z-test, if σ is known and data comes from normal
- TS $= \frac{\bar{X}-\mu_0}{S/\sqrt{n}}$, t-test, $d.f. = n-1$, if data comes from normal
- TS $= \frac{\bar{X}-\mu_0}{S/\sqrt{n}}$, z-test if n is large

Testing Hypotheses about Difference between Two Means

Suppose we test

$$H_0 : \mu_1 - \mu_2 = \Delta_0 \text{ vs. } H_1 : \mu_1 - \mu_2 \neq \Delta_0.$$

Then one can use the following *test statistics* (TS):

- $\text{TS} = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$, z-test, if σ_1 and σ_2 are known and data comes from normal distributions
- $\text{TS} = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$, t-test, $d.f. \cong \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1}\left(\frac{S_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1}\left(\frac{S_2^2}{n_2}\right)^2}$, if data comes from normal distributions
- $\text{TS} = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$, z-test if n_1, n_2 are large